

Which national parks and activity types generate the highest revenue? Are there any trends or anomalies in the 2024 data that should be considered when allocating staff and resources for 2025?

You are a Data Technician Apprentice working in the Data Insights team at **Natural England**, the government body responsible for England's national parks. You have been handed a pre-built aggregated dataset covering visitor numbers, revenue, activity types and satisfaction scores across six national parks for the 2024 calendar year. Your manager has asked you to validate the data, summarise the key figures, build a dashboard, and present your findings before the 2025 resource planning meeting.

File provided: national_park_visitors_aggregated.xlsx

Task 1 Validate

- Open national_park_visitors_aggregated.xlsx and review the structure — check that all columns are present, all rows contain data, and that data types look appropriate.
- Cross-check data integrity: verify row counts, check for blank cells using ISBLANK or conditional formatting, and confirm that no expected columns are missing.
- Apply conditional formatting to highlight potential outliers — flag values in Visitor Count, Revenue, Avg Spend Per Visitor and Satisfaction Score that fall outside a reasonable range.
- Sort and filter the data to focus on specific parks, regions and activity types — identify which subsets are most relevant to the business question.
- Document every issue you find: record what the issue is, where it appears, what you think caused it, and what action you took.

Criteria evidenced: K8 | K10 | K11 | S8 | S9 | S16 | B2 | B4

Task 2 Summarise

- Use SUM, COUNT, AVERAGE, MIN and MAX to calculate overall totals and statistics for Visitor Count, Revenue and Satisfaction Score across the full dataset.
- Use COUNTIF and SUMIF to break down visitor counts and revenue by Park Name, Region and Activity Type.
- Use AVERAGEIF to calculate the average satisfaction score for each park and each activity type.
- Use COUNTIFS, SUMIFS and AVERAGEIFS to answer multi-condition questions — for example, total revenue from Cycling activities in the North region.
- Ensure all your formula results link directly back to the data — do not type calculated values by hand.

Criteria evidenced: K7 | K9 | S7 | B3 | B4

Task 3 Pivot

- Insert a Pivot Table summarising the data by at least two dimensions — for example, Park Name by Activity Type, or Region by Visit Month.

- Add a second Pivot Table or modify the first to show trends across Visit Month — use this to identify seasonal patterns in visitor numbers or revenue.
- Insert a Pivot Chart from your Pivot Table. Choose an appropriate chart type: a bar chart for comparing parks or activities, a line chart for trends over time.
- Format your Pivot Chart with a title, axis labels and a legend. Remove any gridlines or chart elements that add clutter without adding meaning.

Criteria evidenced: K9 | S7 | S8 | B3 | B4

Task 4 Present

- Create a new sheet named Dashboard. This sheet should be clean — no raw data, no visible formulas.
- Copy or link your Pivot Chart and at least two key summary figures (e.g. total annual revenue, highest-scoring park) onto the Dashboard sheet.
- Add at least one Slicer connected to your Pivot Table to allow filtering by Park Name or Region.
- Format the dashboard for a non-technical audience: clear headings, consistent colours, no unnecessary borders or clutter.

Criteria evidenced: K7 | S7 | B2 | B3 | B4

Task 5 Interpret

- Write a structured plain-English summary of your findings. Aim for 200–300 words.
- Identify and describe at least two trends you observed in the data — for example, seasonal patterns, differences between regions, or activity types with consistently high or low performance.
- Identify and describe at least one anomaly you found during validation. State clearly what you think it is, why it is a problem, and what action you recommend.
- State clearly what the data supports and what it does not — identify any limitations or gaps that would affect how confident you are in your conclusions.

Criteria evidenced: K10 | K11 | S9 | B2 | B3

Extension Tasks (Optional — for stronger learners)

- **Median vs Mean:** Calculate the median visitor count and revenue using the MEDIAN function. Compare these to the AVERAGE. Write two sentences explaining what the difference tells you about the distribution of the data and whether outliers are present.
- **Trendline:** Add a trendline to your monthly Pivot Chart. Describe in one or two sentences what the trendline suggests about visitor numbers or revenue going into 2025.
- **Macro:** Record a simple macro that applies your conditional formatting rules automatically. Note what the macro does in plain English and why automating this step saves time when working with larger or regularly updated datasets.

Reflection — answer these questions in writing

1. What validation steps did you carry out, and why are they important before analysis begins?

2. What anomalies did you find in the dataset, and what decisions did you make about each one?

3. How confident are you in your findings, and what additional information or data would increase that confidence?